Lab 7: Fuzzing for Input Validation Bugs

Name	Aryaman D K
SRN	PES1UG24AM344
Assigned repo	L6_L7_39_password_policy (lab7)
Application	password_policy.py - Password Policy Checker
Tools	Python 3, Hypothesis 6, pytest

Objective

To use property-based fuzz testing with the Hypothesis library to find input-validation bugs in **password_policy.py**, and then harden the code so that it handles every unexpected input safely.

Functions under test

first_character_type(pw) - classifies the first character as digit / letter / symbol.
split_credentials(text) - splits a *user:password* string into a (user, password) tuple.
digit_ratio(pw) - returns the fraction of characters in the password that are digits.

Step 1: Installing dependencies

`pip install hypothesis` and `pip install pytest` were run to install the fuzzing library and the test runner.

Step 2: Running the buggy application

Command: `python processor.py` (named `password_policy.py` in this repo). With normal input the program works, but empty or badly formatted input makes it crash.

```
PS C:\Users\Aryaman D K\lab7_buggy\lab7> python password_policy.py
==== Running first_character_type ====
Enter a password:
Traceback (most recent call last):
  File "C:\Users\Aryaman D K\lab7_buggy\lab7\password_policy.py", line 33, in <module>
    main()
  File "C:\Users\Aryaman D K\lab7_buggy\lab7\password_policy.py", line 19, in main
    result = first_character_type(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Aryaman D K\lab7_buggy\lab7\password_policy.py", line 2, in first_character_type
    ch = pw[0]
         ~~^^^
IndexError: string index out of range
PS C:\Users\Aryaman D K\lab7_buggy\lab7>
```

Fig 2.1 - Buggy application crashes on an empty password (IndexError)

Step 3: Completing test_password_policy.py with Hypothesis

The three starter tests were completed so that Hypothesis feeds random Unicode strings and **None** to each function (`st.text() | st.none()`) and asserts that no exception is raised. Three extra property tests were added to check that the outputs are also correct:

Test	Property checked
`test_first_character_type_no_crash`	never raises for any text or None
`test_first_character_type_valid_label`	non-empty text always gives digit / letter / symbol
`test_split_credentials_no_crash`	never raises for any text or None
`test_split_credentials_round_trip`	'user:password' splits back correctly, even if password has ':'
`test_digit_ratio_no_crash`	never raises for any text or None
`test_digit_ratio_in_range`	result is always between 0.0 and 1.0

Step 4: Running the fuzz tests on the buggy code

Command: `python -m pytest test_password_policy.py -v --tb=no`. Result: **5 failed, 1 passed**. Hypothesis automatically shrank each failure to the smallest failing input (listed in the bug table below).

```
PS C:\Users\Aryaman D K\lab7_buggy\lab7> python -m pytest test_password_policy.py -v --tb=no
============================================= test session starts =============================================
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\Aryaman D K\AppData\Local\Programs\Python\Python
311\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Aryaman D K\lab7_buggy\lab7
plugins: anyio-4.15.1, hypothesis-6.170.15, langsmith-0.14.4, locust-2.46.6, platformdirs-4.13.0
collected 6 items

test_password_policy.py::test_first_character_type_no_crash FAILED                                      [ 16%]
test_password_policy.py::test_first_character_type_valid_label PASSED                                   [ 33%]
test_password_policy.py::test_split_credentials_no_crash FAILED                                         [ 50%]
test_password_policy.py::test_split_credentials_round_trip FAILED                                       [ 66%]
test_password_policy.py::test_digit_ratio_no_crash FAILED                                               [ 83%]
test_password_policy.py::test_digit_ratio_in_range FAILED                                               [100%]

=================================================== short test summary info ===================================================
FAILED test_password_policy.py::test_first_character_type_no_crash - ExceptionGroup: Hypothesis found 2 distinct failure
s. (2 sub-exceptions)
FAILED test_password_policy.py::test_split_credentials_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures.
(2 sub-exceptions)
FAILED test_password_policy.py::test_split_credentials_round_trip - ValueError: too many values to unpack (expected 2)
FAILED test_password_policy.py::test_digit_ratio_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub
-exceptions)
FAILED test_password_policy.py::test_digit_ratio_in_range - ZeroDivisionError: division by zero
================================================ 5 failed, 1 passed in 3.59s =================================================
PS C:\Users\Aryaman D K\lab7_buggy\lab7>
```

Fig 4.1 - Fuzz tests on the buggy code: 5 failed, 1 passed

Bugs found by fuzzing

#	Function	Failing input	Error	Fix applied
1	first_character_type	''	IndexError	return 'empty'
2	first_character_type	None	TypeError	return 'invalid'
3	split_credentials	'' / 'alice'	ValueError	return None if ':' missing
4	split_credentials	None	AttributeError	return None for non-string
5	split_credentials	'0::'	ValueError (too many)	split(':', 1)
6	split_credentials	':' / 'user:'	returns empty parts	return None if a part is empty
7	digit_ratio	''	ZeroDivisionError	return 0.0
8	digit_ratio	None	TypeError	return 0.0

Bug 6 was not a crash, but returning an empty username or password is invalid output, so it was hardened at the same time.

Step 5: Fixing the application

Each function now checks that the input is a string and is not empty before using it. **split_credentials** splits only on the first ':' so passwords containing ':' work, and returns **None** for badly formatted input. **main()** prints a clear error message instead of crashing.

```
PS C:\Users\Aryaman D K\SE_LAB_PES1UG24AM344\Lab7_AM344> python password_policy.py
==== Running first_character_type ====
Enter a password: Secret1
First Character Type: letter

==== Running split_credentials ====
Enter user:password (e.g. alice:Secret1): alice:Secret1
User and Password: ('alice', 'Secret1')

==== Running digit_ratio ====
Enter a password to measure: abc123
Digit Ratio: 0.5
PS C:\Users\Aryaman D K\SE_LAB_PES1UG24AM344\Lab7_AM344>
```

Fig 5.1 - Fixed application runs correctly

Step 6: Re-running the fuzz tests on the fixed code

Command: `python -m pytest test_password_policy.py -v`. Result: **all 6 tests passed**.

```
PS C:\Users\Aryaman D K\SE_LAB_PES1UG24AM344\Lab7_AM344> python -m pytest test_password_policy.py -v
============================================= test session starts =============================================
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\Aryaman D K\AppData\Local\Programs\Python\Python
311\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Aryaman D K\SE_LAB_PES1UG24AM344\Lab7_AM344
plugins: anyio-4.15.1, hypothesis-6.170.15, langsmith-0.14.4, locust-2.46.6, platformdirs-4.13.0
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test_password_policy.py::test_first_character_type_no_crash PASSED                                    [ 16%]
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test_password_policy.py::test_split_credentials_no_crash PASSED                                       [ 50%]
test_password_policy.py::test_split_credentials_round_trip PASSED                                     [ 66%]
test_password_policy.py::test_digit_ratio_no_crash PASSED                                             [ 83%]
test_password_policy.py::test_digit_ratio_in_range PASSED                                             [100%]

============================================== 6 passed in 4.06s ==============================================
PS C:\Users\Aryaman D K\SE_LAB_PES1UG24AM344\Lab7_AM344>
```

Fig 6.1 - All fuzz tests pass on the fixed application

Conclusion

Hand-written tests with normal values did not reveal any problems, but Hypothesis found 8 input-validation bugs within seconds by trying empty strings, None and unusual separators. After adding defensive checks, every function handles unexpected input safely and all fuzz tests pass.